Language Models

Software 1.0

computer code

programs

computer

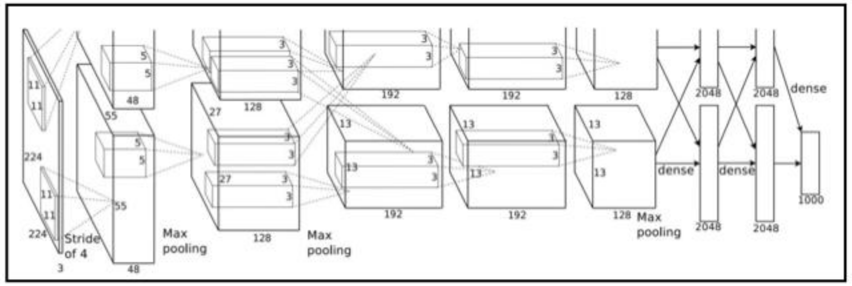
became programmable in **~1940s**

Software 2.0

weights

programs

neural net



fixed function neural net

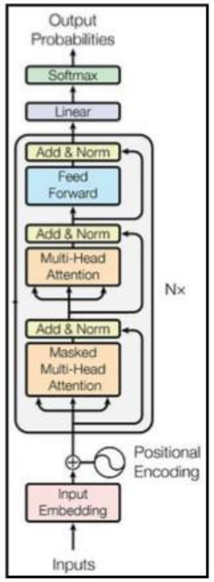
e.g. AlexNet: for image recognition (**~2012**)

Software 3.0

prompts

programs

LLM



~2019

LLM = programmable neural net!

source

Example: Sentiment Classification

Software 1.0

```python
def simple_sentiment(review: str) -> str:
    """Return 'positive' or 'negative' based on a tiny keyword lexicon."""
    positive = {
        "good", "great", "excellent", "amazing", "wonderful", "fantastic",
        "awesome", "loved", "love", "like", "enjoyed", "superb", "delightful"
    }
    negative = {
        "bad", "terrible", "awful", "poor", "boring", "hate", "hated",
        "dislike", "worst", "dull", "disappointing", "mediocre"
    }

    score = 0
    for word in review.lower().split():
        w = word.strip(".,!?;:")          # crude token clean-up
        if w in positive:
            score += 1
        elif w in negative:
            score -= 1

    return "positive" if score >= 0 else "negative"
```

Software 2.0

10,000 positive examples

10,000 negative examples

encoding (e.g. bag of words)

↓ train binary classifier

parameters

Software 3.0

You are a sentiment classifier. For every review that appears between the tags
<REVIEW> ... </REVIEW>, respond with **exactly one word**, either POSITIVE or NEGATIVE (all-caps, no punctuation, no extra text).

Example 1
<REVIEW>I absolutely loved this film—the characters were engaging and the ending was perfect.</REVIEW>
POSITIVE

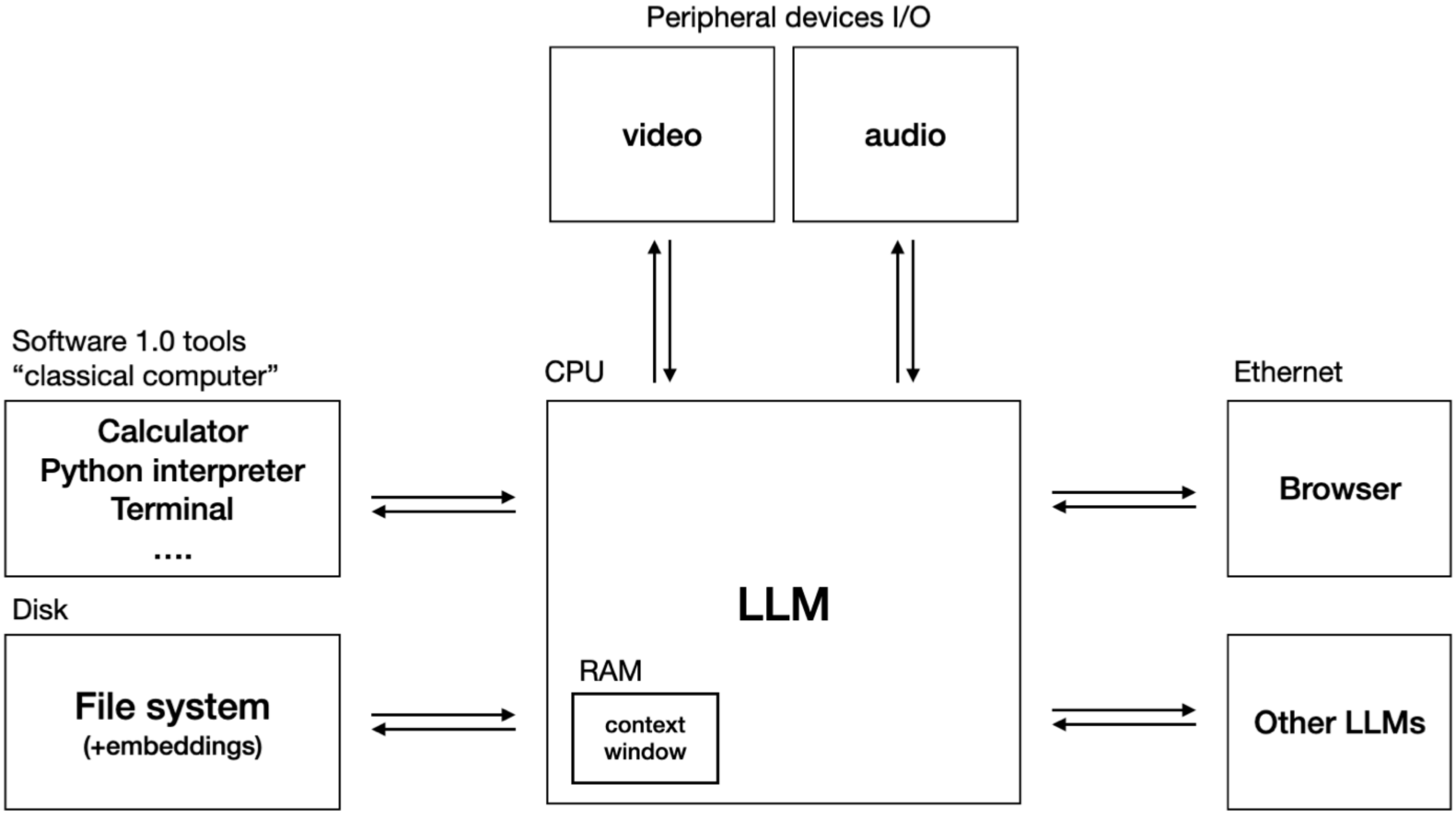
Example 2
<REVIEW>The plot was incoherent and the acting felt forced; I regret watching it.</REVIEW>
NEGATIVE

Example 3
<REVIEW>An energetic soundtrack and solid visuals almost save it, but the story drags and the jokes fall flat.</REVIEW>
NEGATIVE

Now classify the next review.

source

LLM OS



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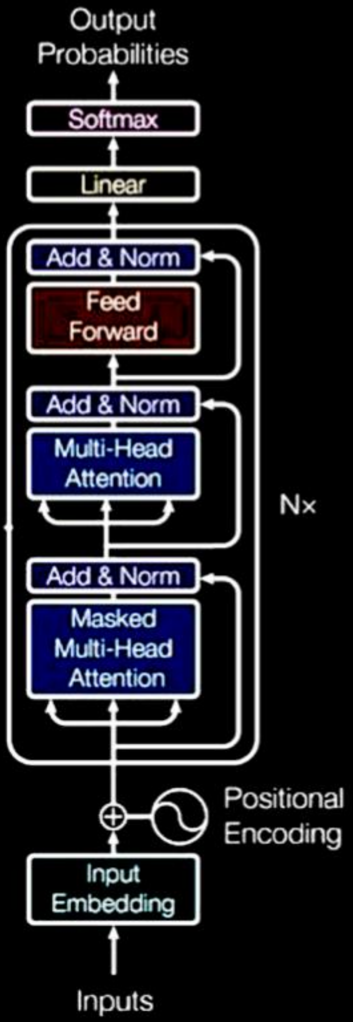
LLMs are "people spirits": stochastic simulations of people.

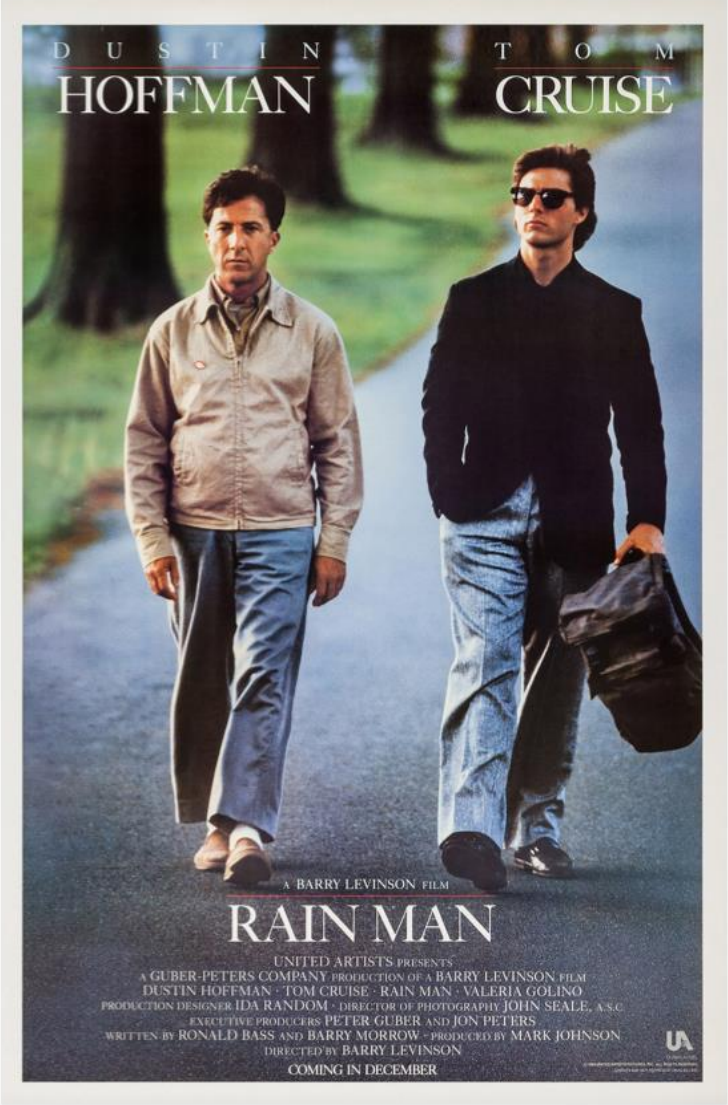
Simulator = autoregressive Transformer



=> They have a kind of emergent "psychology".

source

Encyclopedic knowledge/memory, ...



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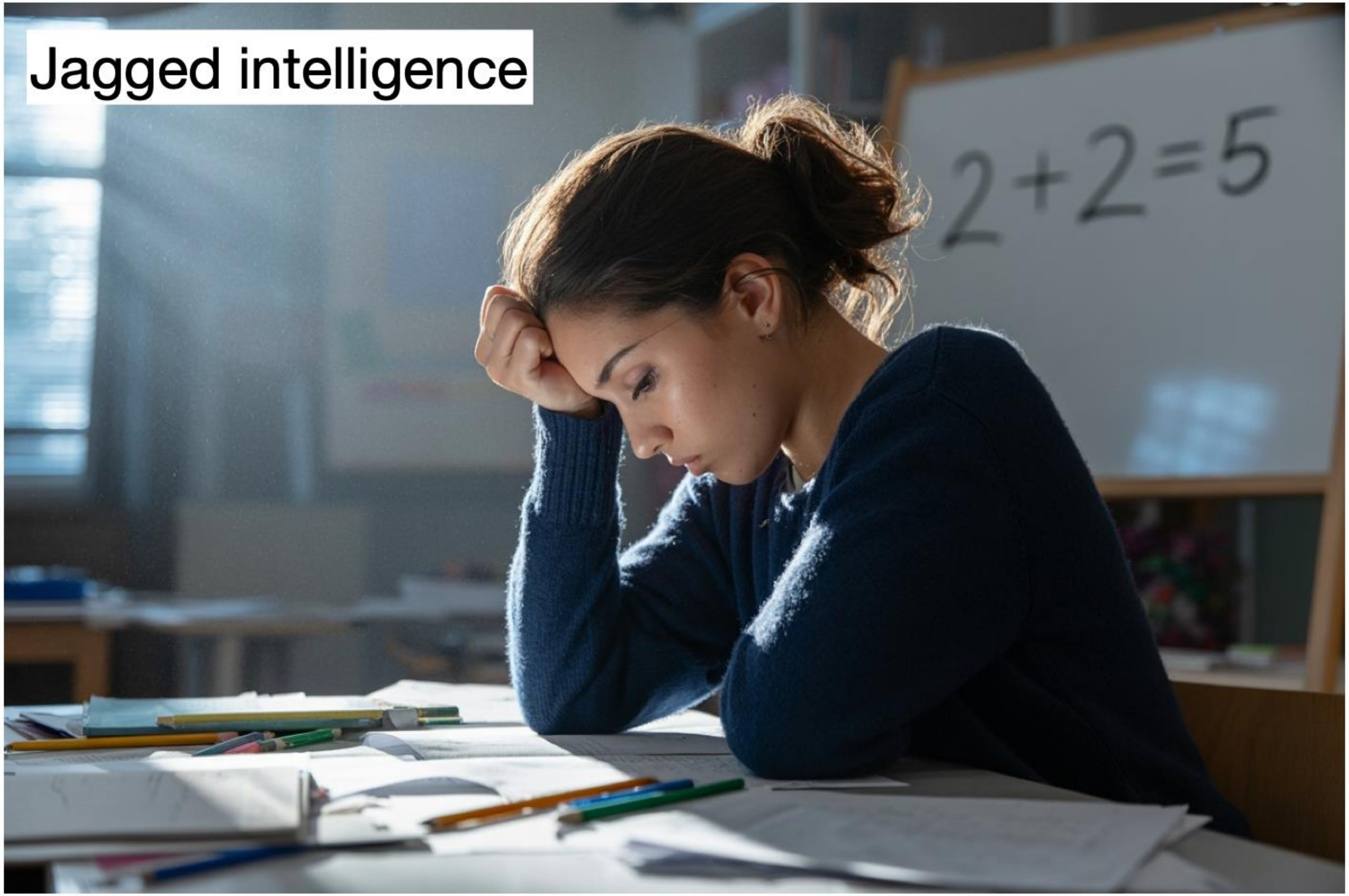
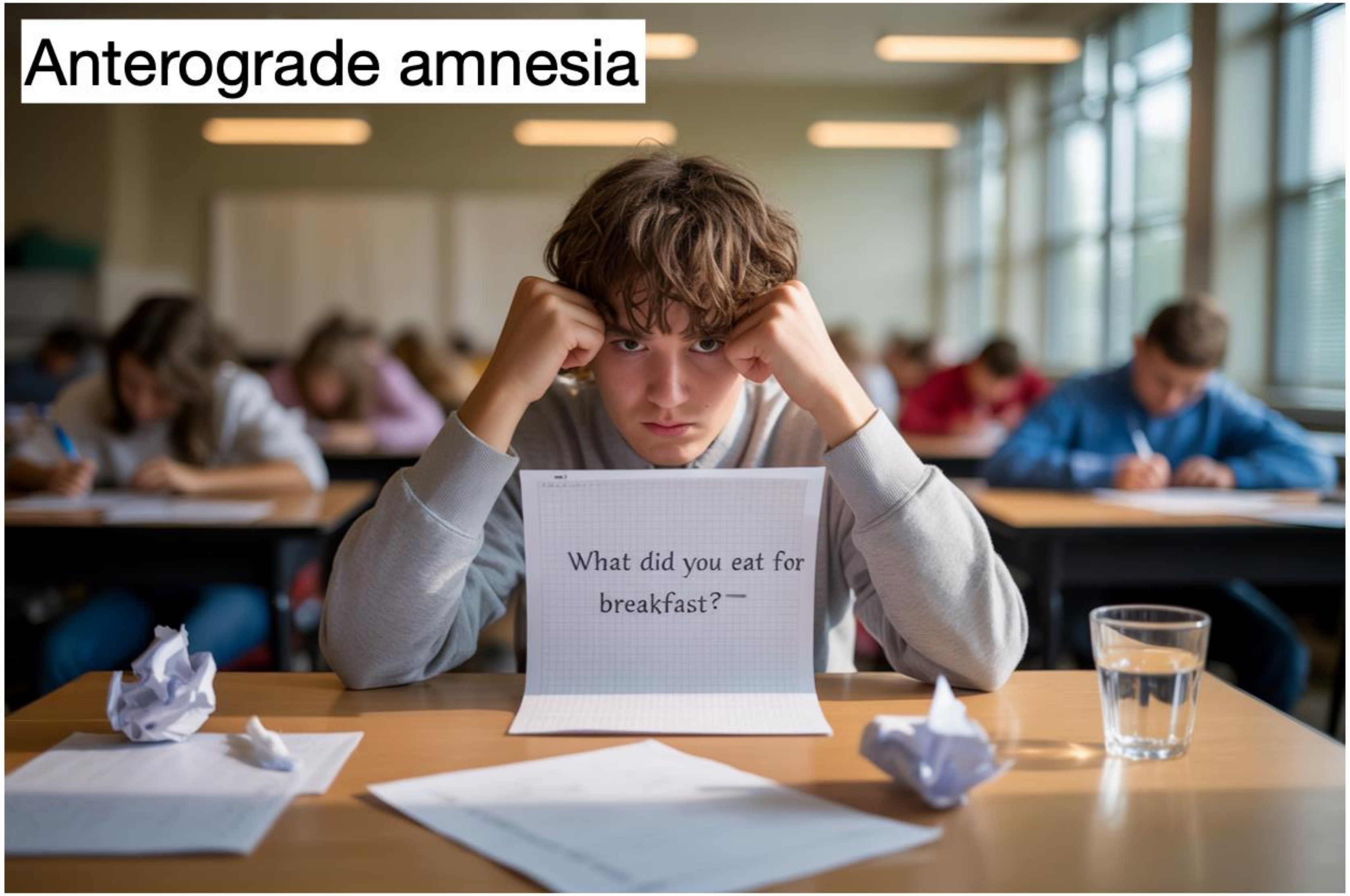


Famous examples: 9.11 > 9.9, two 'r' in 'strawberry', ...

source



Context windows ~= working memory.

No continual learning, no equivalent of "sleep" to consolidate knowledge, insight or expertise into weights.

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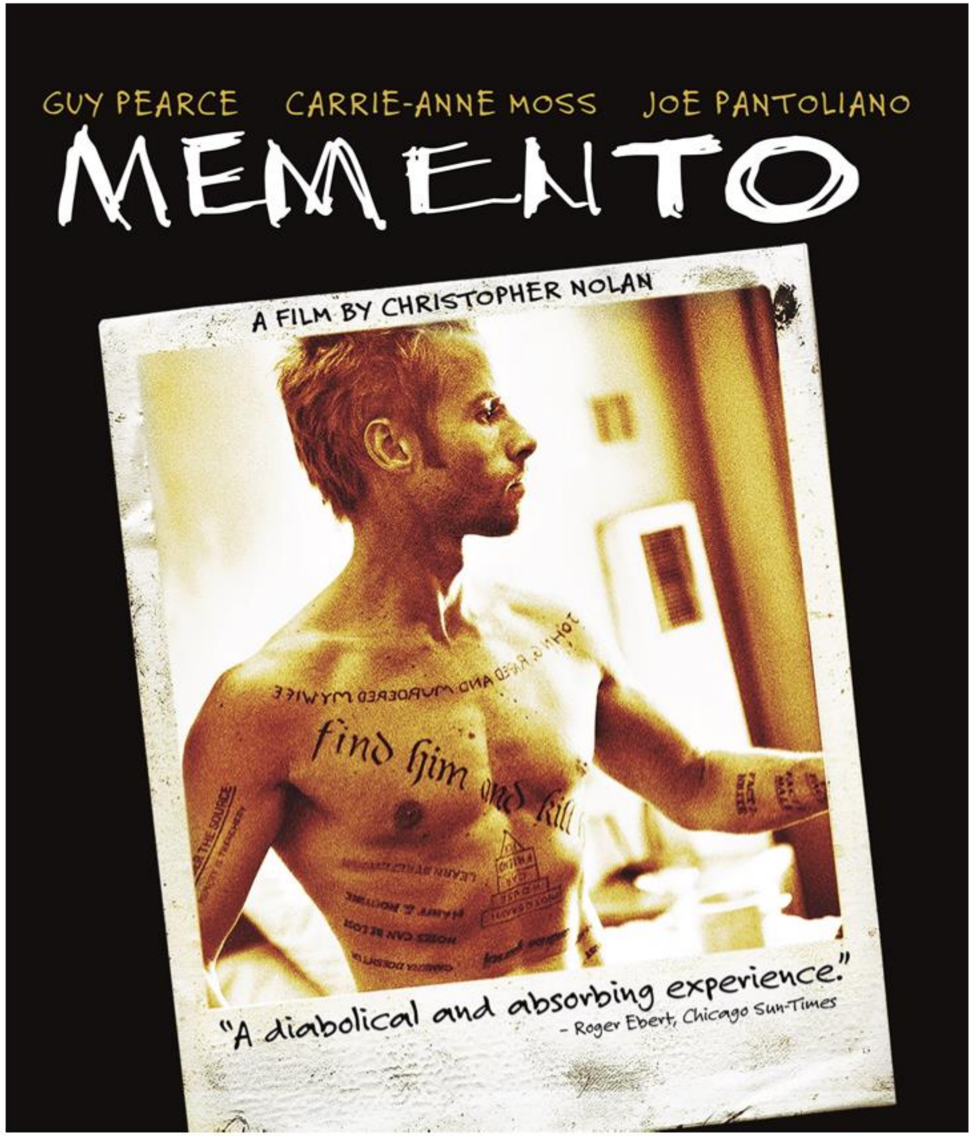
In popular culture...



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=> Prompt injection risks, e.g. of private data

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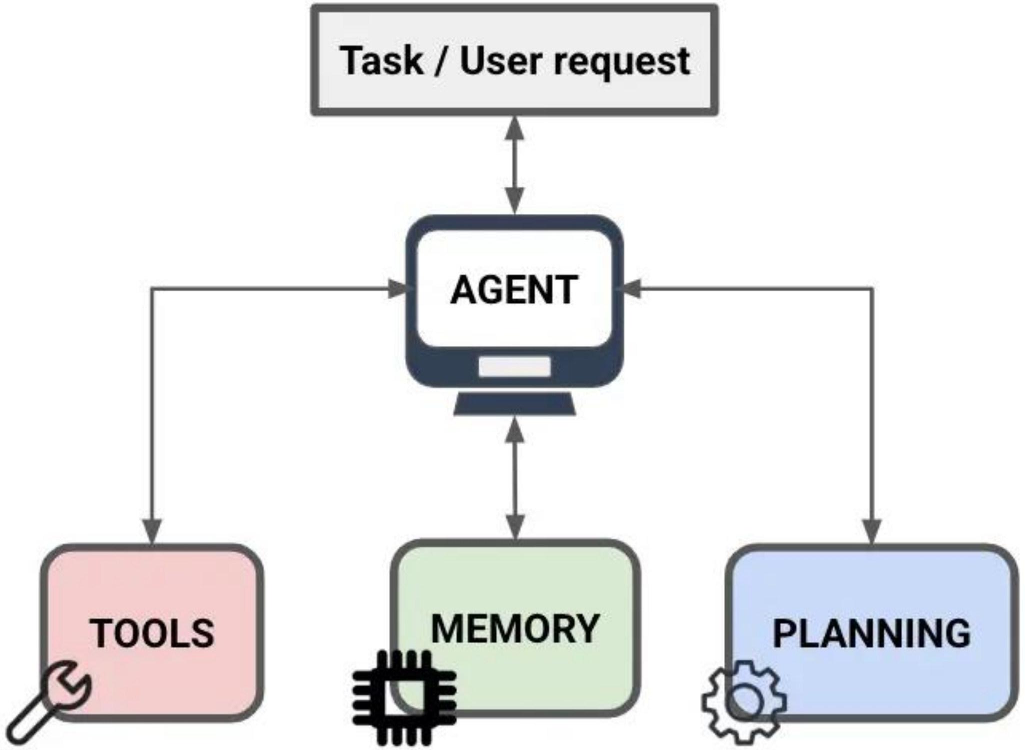
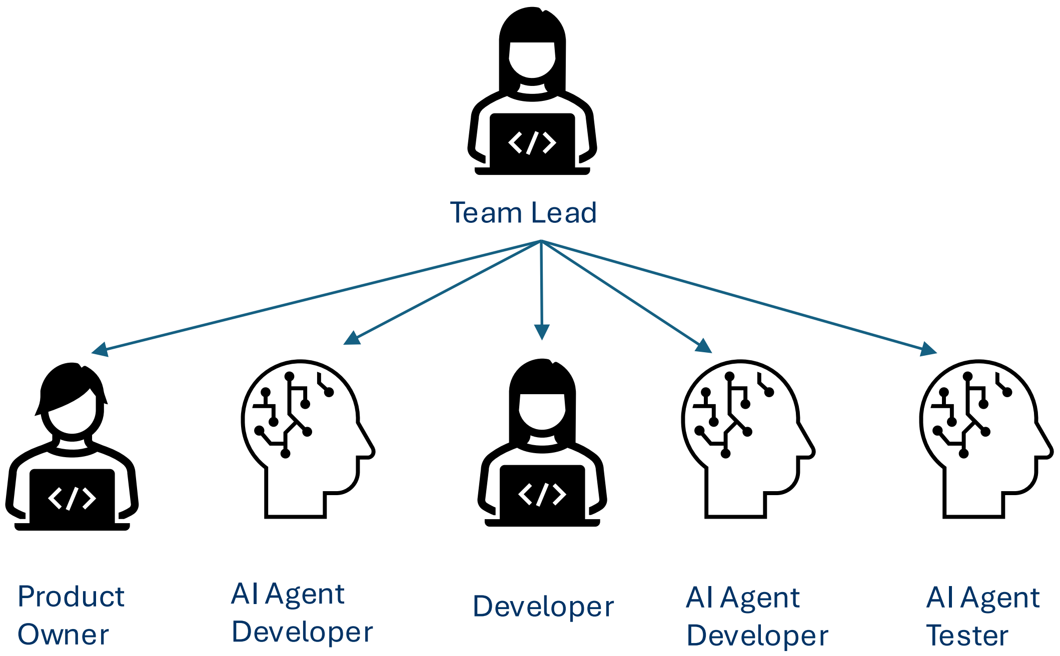


digital workers:



Then let's find out what's behind it ...